Compile Findings, Refresh Plan & Handover

Appointment Volume Forecasting

1. Summary of Results

The Nexora Care Flow forecasting analysis evaluated six approaches for predicting daily appointment volume using historical appointment data, weekly patterns, and holiday information. The objective was to support staffing, appointment capacity planning, workload management, and resource allocation. The analysis used 675 observations for training and the final 56 observations for testing.

The principal finding is that SARIMAX is the strongest-performing model among the six approaches evaluated.

Rank	Model	MAE	RMSE
1	SARIMAX	33.09	44.12
2	XGBoost	49.22	62.31
3	Seasonal Naive	51.46	95.35
4	Prophet	56.41	81.42
5	ARIMA	110.85	119.31
6	Naive baseline	113.34	152.89

Lower MAE and RMSE indicate better forecasting performance.

Key Findings

- Weekly seasonality matters: the Seasonal Naive model substantially outperformed the ordinary Naive baseline.
- SARIMAX performed best: its weekly seasonal component and calendar predictors helped capture demand patterns.
- Holiday effects are important: the fitted SARIMAX model estimated a negative holiday association with appointment volume.
- XGBoost is a credible alternative: it achieved the second-lowest MAE and RMSE using is_holiday and day_of_week.
- Basic ARIMA was insufficient: without an explicit weekly seasonal component, it struggled to capture weekday-to-weekend changes.

2. Recommended Operational Decision

Adopt SARIMAX(1,1,1) × (1,1,1,7) as the provisional primary forecasting model, retain XGBoost as a challenger, and keep Seasonal Naive as the simple benchmark.

Use forecasts as decision support rather than automatic staffing or appointment-allocation instructions. Managers should consider clinical requirements, staff availability, and unusual events before making operational decisions.

3. Limitations and Risks

Limitation	Implication	Recommended Action
Approximately two years of historical data	Longer-term trends may not be fully represented.	Collect more historical observations as they become available.
Limited predictors	Other operational factors may affect demand.	Consider additional features in a separately evaluated future iteration.
One configuration per model	Best possible performance for each model has not been established.	Tune parameters and compare alternatives.
One 56-day test period	Performance may vary across seasons or demand conditions.	Conduct rolling-origin or expanding-window validation.
Holiday coefficient reflects association, not causation	Other factors may contribute to lower demand on holidays.	Validate the effect across additional periods and operational records.
No demonstrated live deployment	Test accuracy does not guarantee production performance.	Pilot the model before relying on it operationally.

Interpretation: MAE of 33.09 means the average absolute forecast error was approximately 33 appointments per day on the evaluated test period. It does not guarantee that future forecasts will be this accurate.

4. Monitoring and Model Refresh Plan

Step 1 — Update the data

Suggested frequency: Daily or whenever the latest appointment records become available.

Append new records, validate dates, check missing values, and calculate daily appointment volume and the existing calendar variables. Confirm dates are continuous and correctly ordered.

Step 2 — Monitor forecast accuracy

Suggested frequency: Weekly, once actual appointment counts are available.

Compare forecasts with actual volumes using MAE and RMSE. Track errors by weekday and holiday status to identify underperformance.

Step 3 — Flag deterioration

Suggested frequency: At each monitoring review.

Investigate sustained increases in forecast errors, unexpected demand changes, missing data, or changes in weekly patterns. Treat these as investigation triggers, not automatic proof of model failure.

Step 4 — Retrain and validate

Suggested frequency: Monthly initially, or sooner if performance deteriorates.

Refit SARIMAX on expanded historical data. Re-evaluate against XGBoost and Seasonal Naive using the same chronological validation procedure. Change the production model only when evidence supports it.

Step 5 — Record and communicate

Suggested frequency: At every refresh.

Record the dataset end date, model specification, validation results, forecast horizon, model version, and decisions made. Share forecast outputs and material changes with operational managers.

Suggested Alert Policy

- Warning: MAE or RMSE rises materially above the established baseline.
- Investigation: deterioration persists across consecutive review periods or particular weekdays.
- Refresh decision: retrain and compare models using a fresh chronological validation window.
- Rollback: retain the previous approved model until a replacement passes validation.

Set numerical alert thresholds after sufficient live observations have accumulated.

5. Handover Checklist

- **Data pipeline:** Confirm the processed CSV contains the date, appointment_volume, is_holiday and day_of_week.
- **Reproducible training:** Document the SARIMAX specification, training period, package versions and preprocessing steps.
- **Validation:** Preserve original holdout results and add rolling-origin validation before production adoption.
- **Forecast output:** Save forecast dates, predicted volumes and appropriate uncertainty intervals where available.
- **Monitoring:** Maintain a dated log of actuals, forecasts, MAE, RMSE and data-quality issues.
- **Ownership:** Assign responsibility for data updates, model refreshes, approval and operational communication.

6. Final Handover Statement

Nexora Care Flow's initial forecasting evaluation identified SARIMAX as the best-performing of six tested approaches, achieving MAE of 33.09 and RMSE of 44.12 on the final 56-day test period. The results support further development of a forecasting process that accounts for weekly seasonality, calendar effects, and historical demand patterns.

The next phase should focus on validating results across multiple time periods, establishing a repeatable data-update pipeline, monitoring forecast errors, and defining controlled model-refresh procedures. XGBoost and Seasonal Naive should remain comparison models. Operational adoption should follow a monitored pilot, with managers retaining responsibility for final staffing and capacity decisions.